

Experiments with Theta Cycles Modulo p^2 : Hasse-Jet Duality and a Shifted-Conic Conjecture

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Abstract

Some weights in theta cycles modulo p^2 are still unknown. We study the next unresolved wedge by exact computation. In 692 ordinary cross-wedge cells, one shifted quadratic matches all 458 interior obstructions. The 234 disagreements lie on three affine boundary lines. Two separate tests give 2,324 exact matches with one factorial boundary scalar. We also prove that the obstruction is a first Hasse jet. The relevant Hasse quotients have a perfect pairing, and theta is skew-adjoint for this pairing. We give the minimal-weight projection and prove that exact filtration separates the boundary directions. These theorems reduce the proposed shifted-conic law to one factorial block identity and two endpoint identities. Those three identities remain conjectures. The paper includes code, negative controls, and exact tests that can detect a failure.

Keywords: theta cycles; modular forms modulo prime powers; experimental mathematics; Hasse invariant; Bockstein class; weight filtration.

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1 Introduction

The theta cycle of a modular form modulo a prime is classical. Modulo p^2 , weight lowering has a new first-order obstruction. A form may lower in weight modulo p but fail to lower in the same way modulo p^2 . Ahlgren, Raum, and Richter determined the initial segment and many later parts of the cycle [1]. Some low points in the next band remain unknown for general primes.

The basic filtration theory modulo p comes from work of Swinnerton-Dyer and Serre [10, 9]. Katz gave a geometric theory of modular forms in characteristic p , including the Hasse invariant and the theta operator [7]. Later work on modular weights and companion forms developed this viewpoint further [5, 4].

We ask one exact question. Does the quadratic pattern in the resolved wedge remain valid after reflection into the next wedge? The first candidate failed on every tested $D = 0$ cell. We then added the missing minimal-weight projection. In the interior, the lower Bockstein class was controlled by

$$i'^2 + (k - 1)i' - (n + 1)(n + 2) \quad \text{in } \mathbb{F}_p. \quad (1)$$

All disagreements occurred on three affine lines. Their full residuals then suggested one factorial block and two triangular endpoint formulas.

The paper explains how we found and tested this formula. The calculation uses exact modular arithmetic and Sturm-safe remainders. It includes a failed baseline, a blind prime extension, and a separate test in a two-dimensional cusp-form space. We then prove the geometry used by the experiment. The weight-lowering obstruction is a first Hasse jet. The ambient quotient has a perfect pairing, and theta is skew-adjoint. Multiplication by the integral Hasse inclusion commutes with theta modulo p^2 . We also give the minimal-weight projection. Exact filtration separates the proposed boundary directions.

Three identities remain open. The first-Hasse-jet formula should reduce to a factorial block identity. This identity would give the generic shifted-conic law and the observed scalar on the third line. Two endpoint coefficients remain open on the other boundary lines. We state all three claims as conjectures because the present evidence is finite.

The experiments identify the classification problem and the three boundary cases. The theorems explain why the candidate has this form. They also reduce the open part to explicit identities.

1.1 Main results and limits

The paper has three kinds of results.

1. **Uniform theorems.** We prove the first-Hasse-jet description, perfect Hasse-quotient duality, theta skew-adjointness, the explicit minimal-weight projection, and separation of the boundary directions.
2. **Exact finite results.** The main census has 692 cells. Two later tests give 2,324 exact matches with the proposed third-line scalar. A separate index audit checks 210,176 parameter cells through $p = 101$.
3. **Open claims.** One factorial block identity and two endpoint identities are still conjectural. The full shifted-conic law is conditional on these identities.

This split is used throughout the paper. No finite calculation is stated as a theorem for all primes.

1.2 Relation to prior work

Ahlgren–Raum–Richter [1] prove the initial segment, further exact values or bounds modulo p^2 , and a quadratic condition for the exceptional positions in their resolved region. Our code also checks the resolved $D = 2$ subwedge, whose conic is

$$i'^2 + (k - 1)i' - n(n + 1) = 0.$$

We study the complementary $D = 0$ subwedge. Reflection changes the constant term to $(n + 1)(n + 2)$. A new first-Hasse-jet projection is needed before this expression gives the actual obstruction. This is not a restatement of Ahlgren–Raum–Richter. The geometric pairing alone also does not prove the shifted-conic law.

1.3 Guide to the paper

Section 2 gives the finite experiment, negative controls, validation tests, and failure rules. The next two sections give the first-Hasse-jet and direct block calculations. Section 7 separates the finite evidence from the open claims. The final sections state the conditional shifted-conic consequences, summarize theorem status, and give exact reproduction commands.

2 Experimental design and discovery path

2.1 Objects, arithmetic, and inclusion rules

The experiment concerns the filtration of $\theta^i f$ modulo p^2 for ordinary level-one modular forms of weight $4 \leq k < p - 1$. Every reported comparison uses exact integer arithmetic. Fourier expansions are retained past the Sturm bound at each relevant weight, so equality of the retained series is an exact equality of modular forms in the tested case. No floating-point approximation, numerical tolerance, random sampling, or model fitting is used.

The computational record contains 5,572 cells. Of these, 5,434 satisfy the ordinary condition $k < p - 1$. The main experiment uses the 692 ordinary, nondegenerate cells for which a $D = 0$ index and its Tate complement lie in the proved $D = 2$ wedge. These cells range over primes $17 \leq p \leq 43$ and weights $8 \leq k \leq 22$. A cell is included by these arithmetic conditions alone, before its obstruction class or the value of the candidate quadratic is inspected.

For each cell, the program performs four exact operations:

1. reduce $\theta^i f$ modulo p^2 against an integral echelon basis of M_{w-} ;
2. divide the resulting canonical remainder by p and reduce modulo p , producing the lower Bockstein class $\mathcal{B}_i$;
3. independently construct the shifted leading-block candidate

$$\mathcal{L}_i = q_1(n, i') X^p \theta^{i-p} f, \quad q_1(n, i') = i'^2 + (k-1)i' - (n+1)(n+2);$$

4. compare the complete canonical remainders, not a selected coordinate or a short Fourier prefix.

The program records both agreements and failures. For each failure, it stores the full residual $\mathcal{B}_i - \mathcal{L}_i$ whenever the proposed identity fails.

2.2 How the pattern was found

We found the pattern in stages. The first candidate reused the $D = 2$ formula one Hasse step lower. It failed in all 1,593 ordinary, nondegenerate $D = 0$ cells. This negative control showed that a new first-order correction was needed.

Reflection to the complementary wedge then suggested replacing the earlier quadratic $i'^2 + (k-1)i' - n(n+1)$ by $q_1(n, i')$. On the 692 cross-wedge cells, the resulting class agreed with the direct Bockstein in 458 cases. The 234 disagreements did not scatter: they were exactly the cells on the three affine lines

$$i' = n + 2, \quad i' + n = 2p - k - 1, \quad i' + n = 2p - k.$$

Each line contains 78 cells. Away from these lines, the direct class vanished in exactly 20 cases, and these were exactly the cases $q_1(n, i') = 0$. Thus the shifted conic was found only after the first formula failed and the full residuals identified the three boundary cases.

The boundary residuals suggested further exact formulas. On $i' = n + 2$ the residual always occupied one explicit block direction, and rational reconstruction across primes produced the scalar

$$-m(k - 2m - 2)!, \quad m = p - 2 - n.$$

On the two triangular lines, exact elimination placed every residual in an explicit triangular span and recovered endpoint coefficient $-(m+1)$. These results led to the Hasse-jet and filtration arguments proved below.

2.3 Validation and negative controls

Table 1 separates discovery calculations, blind extensions, independent form tests, and symbolic-index audits. A zero failure count means that the stated exact identity held throughout the specified finite range. It does not convert the identity into a theorem for arbitrary primes.

The weight-24 experiment is important because the relevant cusp-form space is two-dimensional. It tests two generators and their projective linear combinations rather than repeating the calculation on a single one-dimensional family. The third-line extension is also separate from the original 78 boundary cells: it searches new primes and weights after the one-block shape and factorial scalar have been specified.

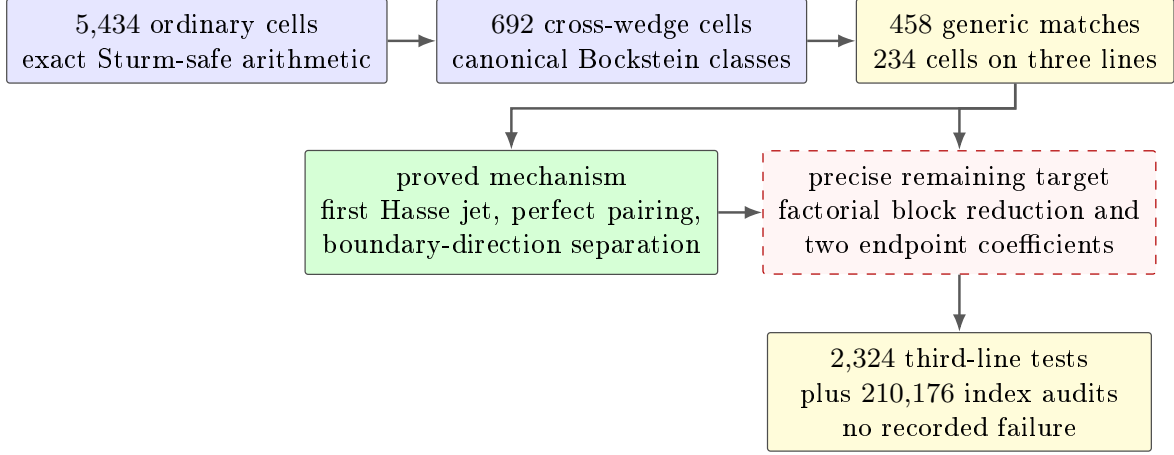


Figure 1: The experiment-to-theorem path. Exact computation first isolated the generic quadratic and its three boundary lines. The green box records the uniform results proved here. The dashed box is the remaining symbolic identity; finite tests support it but do not prove it.

2.4 What would falsify the conjectural mechanism

Each computational claim has a direct failure test. Any of the following would disprove part of the proposed mechanism:

1. an interior cross-wedge cell with $\mathcal{B}_i \neq q_1(n, i')X^p\theta^{i-p}f$;
2. a cell with $q_1(n, i') = 0$ and nonzero interior Bockstein, or an interior zero with $q_1(n, i') \neq 0$;
3. a third-line residual outside the predicted one-block direction;
4. a third-line scalar different from $-m(k - 2m - 2)!$ or dependent on the chosen ordinary form;
5. a triangular residual outside the stated span or with endpoint coefficient different from $-(m + 1)$.

The released scripts report each failure type separately. The tests remain useful even though the factorial block reduction is still open.

3 Notation and proof roadmap

3.1 Core notation

The symbols used repeatedly in the proof are collected here. The band level $D(i)$ is included explicitly so that the later references to the $D = 0$ and $D = 2$ strata are self-contained.

Audit	Exact range	Outcome
Ordinary-cell census	5,434 cells	Establishes the ambient finite record and its admissible subfamilies.
Naive lower-block control	1,593 $D = 0$ cells	The reused $D = 2$ formula failed in all cells, isolating the missing projection term.
Cross-wedge discovery	692 cells	458 interior matches, 234 boundary disagreements, and 20 shifted-conic zeros.
Explicit first-Hasse-jet lift	The same 692 cells	No weight, mod- p , or canonical-Bockstein mismatch.
Third-line blind extension	464 cells through $p = 61$	No zero residual, block-shape failure, or factorial-scalar mismatch.
Independent weight-24 forms	186 ordinary forms and 1,860 tests through $p = 43$	No zero residual, block-shape failure, or scalar mismatch.
Triangular endpoints	156 boundary cells	Endpoint rank increased by one and the coefficient was $-(m + 1)$ in every case.
Index and filtration audit	210,176 parameter cells through $p = 101$	No exponent, weight, pairing, or filtration-gap failure.

Table 1: Exact audits used in the discovery and validation of the proposed shifted-conic mechanism. The two large validation experiments were written as falsification tests: they count zero residuals, shape failures, and scalar mismatches separately.

Symbol	Meaning
p, k, f	$p \geq 5$ is prime; f is an ordinary level-one form of weight $4 \leq k < p - 1$.
θ, ω_{p^a}	The operator $q d/dq$ and weight filtration modulo p^a , respectively.
$\delta(\theta^i f)$	The integer defined on the admissible cells by $\omega_{p^2}(\theta^i f) = k + 2i + \delta(\theta^i f)P$.
$i = np + i'$	Band index, with n the segment number and i' the position inside the segment.
P	The full Hasse drop $p(p - 1)$.
$D(i)$	The least integer D such that $k + 2i + DP \geq \omega_p(\theta^{i-p} f) + p(p + 1)$; this is the inherited band upper-bound level.
w_-	The candidate lower weight $k + 2i - P$ in the $D = 0$ stratum.
$A, \tilde{A}, S, Z$	A is the Hasse invariant over $\mathbb{F}_p$, $\tilde{A} = E_{p-1}$ is its chosen integral lift, $S = \text{div}(A)$, and $Z = pS$ is its length- p thickening. In a purely integral formula, A may be used for E_{p-1} after this is stated explicitly.
Q_r	The Hasse quotient $H^0(Z, \omega^r _Z)$. Restriction gives an injection $M_r(\mathbb{F}_p)/A^p M_{r-p}(\mathbb{F}_p) \hookrightarrow Q_r$; it is an isomorphism when the restriction map is onto.
$\mathcal{B}_i, \mathcal{L}_i$	The Bockstein class of $\theta^i f$ at weight w_- and its shifted leading-block candidate. Their difference is the boundary residual.
X	The divided Eisenstein series $E_2/12$ modulo p ; powers of X form the direct block basis used in the reduction.
α, Φ_r, η_r	The first Hasse jet $(A - 1)/p$, the periodicity quotient $(\theta^{r+p-1} f - \theta^r f)/p$, and the lift error $(G_r - \theta^r f)/p$, all reduced modulo p .
Z_d	The block direction $X^d \theta^{p-k} f$ in the factorial reduction.
$i^\dagger$	The complementary index $p^2 + 1 - k - i$ used in the Tate-dual comparison.

Terminology. A *first Hasse jet* is a class on the thickened Hasse divisor Z . A *Bockstein class* is the intrinsic obstruction to an integral weight-lowering lift. A *minimal-weight projection* is a chosen characteristic-zero representative of the smallest candidate weight. The *block basis* is the direct basis built from powers of X and theta iterates. A *boundary residual* is the difference between the Bockstein class and its shifted leading-block candidate on one of the three affine boundary lines.

An *admissible* $D = 0$ - $D = 2$ *cross-wedge cell* means that the lower index has $D(i) = 0$, its Tate complement has $D(i^\dagger) = 2$, and there are integers m, d such that

$$n = p - 2 - m, \quad i' = p - k + m + 1 + d,$$

with

$$1 \leq m \leq \left\lfloor \frac{k-4}{2} \right\rfloor, \quad 0 \leq d \leq k - 2m - 1.$$

These are the cross-wedge inequalities used below. In particular, every factorial and every power in the uniform statements has a nonnegative exponent.

3.2 Dependency roadmap

The proof has five logically separate layers. The first three are uniform in the admissible parameters and do not use the conjectural scalar identity.

Layer	Main result	Logical role
Geometric	Perfect Hasse-quotient duality and theta skew-adjointness	Uses duality on Z ; independent of all finite computations and block formulas.
Integral	Bockstein realization, Hasse inclusion, and the explicit periodic first-Hasse-jet formula	Uses integral Eisenstein lifts and the first ordinary low-point congruence; supplies the minimal-weight projection.
Filtration	Uniform separation of all three boundary directions and nonvanishing of the factorial scalar candidate	Uses exact mod- p filtration; independent of the factorial block reduction.
Computational	Exact classification of 692 cross-wedge cells and 2,324 third-line stress tests	Proved only on the deposited finite range; discovers and falsifies formulas but is not extrapolated to all p .
Conditional	Factorial block reduction and two triangular endpoint coefficients	Would identify the actual boundary residuals and imply the shifted-conic filtration law.

Thus the dependency chain is

$$\boxed{\text{geometry}} + \boxed{\text{integral first Hasse jet}} \implies \boxed{\text{explicit Bockstein class}},$$

$$\boxed{\text{explicit Bockstein class}} \xrightarrow[\text{open}]{\text{block reduction}} \boxed{\text{shifted-conic law}}.$$

The boundary-separation theorem branches from the filtration layer before the open implication and therefore remains unconditional.

4 Bocksteins and Hasse quotients

4.1 The weight-lowering Bockstein

Let $R = \mathbb{Z}_{(p)}$, and let $M_w(R)$ denote the space of level-one modular forms of weight w with coefficients in R . We identify a form with its q -expansion. Write

$$\overline{M}_w := M_w(R)/pM_w(R)$$

and let $\overline{\mathcal{D}}$ be the space of divided congruences modulo p , viewed through their q -expansions.

Definition 4.1. Let $g \in \mathcal{D}(R/p^2R)$ and suppose that $g \pmod{p}$ is represented by some $h \in M_w(R)$. Define

$$\beta_w(g) := \left[\frac{g-h}{p} \right] \in \overline{\mathcal{D}}/\overline{M}_w.$$

Here any lift of the q -expansion of g to R may be used in the numerator.

Lemma 4.2. The class $\beta_w(g)$ is independent of the choices in Definition 4.1. Moreover,

$$\beta_w(g) = 0 \iff g \equiv H \pmod{p^2} \text{ for some } H \in M_w(R).$$

Consequently, if w lies in the admissible weight class of g , then

$$\beta_w(g) = 0 \iff \omega_{p^2}(g) \leq w.$$

Proof. Suppose first that $h, h' \in M_w(R)$ both represent $g \pmod{p}$. Then $h-h'$ has q -expansion divisible by p . The q -expansion lattice $M_w(R)$ is saturated: an integral modular form whose coefficients are all divisible by p belongs to $pM_w(R)$. One can also see this directly by forward substitution in the standard integral echelon basis, whose pivot coefficients are units. Thus $h-h' \in pM_w(R)$. Hence

$$\frac{g-h}{p} - \frac{g-h'}{p} = \frac{h'-h}{p} \in M_w(R),$$

so the two choices define the same class modulo $\overline{M}_w$. Changing the integral lift of g changes the quotient by an arbitrary multiple of $p^2/p = p$, and hence does not change its reduction modulo p . This proves well-definedness.

If $\beta_w(g) = 0$, there is an $m \in M_w(R)$ such that $(g-h)/p \equiv m \pmod{p}$. Thus

$$g \equiv h + pm \pmod{p^2},$$

and $h+pm$ belongs to $M_w(R)$. Conversely, suppose that $g \equiv H \pmod{p^2}$ for some $H \in M_w(R)$. Since $H \equiv h \pmod{p}$, the same echelon-basis argument gives $H-h = pm$ for some $m \in M_w(R)$. It follows that

$$\frac{g-h}{p} \equiv \frac{H-h}{p} = m \pmod{p},$$

so $\beta_w(g) = 0$ in $\overline{\mathcal{D}}/\overline{M}_w$. For the final equivalence, assume that every smaller admissible representative weight differs from w by a nonnegative multiple of $P = p(p-1)$. Multiplication by the corresponding power of $E_{p-1}^p \equiv 1 \pmod{p^2}$ raises such a representative to weight w without changing its reduction modulo p^2 . Therefore a representative of weight at most w exists exactly when one of weight w exists. This is the asserted filtration criterion. $\square$

Remark 4.3. The canonical echelon remainder used in the computations is a coordinate representative of $\beta_w(g)$. Reducing g modulo the integral echelon basis of $M_w(R)$ kills the same leading coefficients as the lift h in the proof. The resulting remainder is divisible by p , and division by p followed by reduction modulo p gives the displayed class. Thus the computational zero test is intrinsic even though its coordinate vector depends on the chosen echelon basis.

4.2 Hasse duality and the Eisenstein bridge

Let $p \geq 5$, put $P = p(p-1)$, and work first on a compactified Katz modular curve X with auxiliary full level 3. Let ω be the Hodge bundle, let C be the cusp divisor, and let $A \in H^0(X, \omega^{p-1})$ be the Hasse invariant. Its zero divisor S is reduced and disjoint from C . Put $Z = pS$, so that Z is the Cartier divisor cut out by A^p . For every integer r define

$$\mathcal{Q}_r := \omega^r / A^p \omega^{r-P} \simeq \omega^r|_Z, \quad Q_r := H^0(Z, \mathcal{Q}_r).$$

All schemes, line bundles, and sections in this subsection are over $\mathbb{F}_p$ unless another base is written explicitly. The constructions below commute with the tame deck group. Its order is prime to p , so the Reynolds operator is exact. The canonical pairing below is deck-invariant, and its restriction to invariant subspaces is perfect. This gives the same statement at level one. Equivalently, one may formulate it directly on the tame compactified moduli stack.

Proposition 4.4 (Perfect Hasse-quotient pairing). *For every r there is a canonical perfect pairing*

$$\langle \ , \ \rangle_r : Q_r \times Q_{P+2-r} \longrightarrow \mathbb{F}_p.$$

It is the product pairing followed by the trace map on the finite Gorenstein scheme Z .

Proof. Because Z is an effective Cartier divisor on the smooth curve X , its dualizing sheaf is

$$\omega_Z \simeq (\omega_X \otimes \mathcal{O}_X(Z))|_Z.$$

Kodaira–Spencer gives $\omega_X(C) \simeq \omega^2$, while $\mathcal{O}_X(Z) \simeq \omega^P$. The canonical section of $\mathcal{O}_X(C)$ is nonzero on Z , since $C \cap Z = \emptyset$. Consequently

$$\omega_Z \simeq \omega^{P+2}|_Z.$$

Multiplication now gives

$$Q_r \otimes Q_{P+2-r} \longrightarrow \omega_Z.$$

Composing global sections with $\text{Tr}_Z : H^0(Z, \omega_Z) \rightarrow \mathbb{F}_p$ gives the displayed pairing. Duality for the finite Gorenstein scheme Z identifies Q_{P+2-r} with $\text{Hom}_Z(Q_r, \omega_Z)$, so the pairing is perfect. For the level-one statement, average a dual vector over the tame deck group. Since the group order is a unit in $\mathbb{F}_p$, this averaging shows that the restriction to invariant subspaces is still perfect. $\square$

Let Θ denote the Katz theta operator in characteristic p [8], normalized by its q -expansion

$$\Theta\left(\sum a_n q^n\right) = \sum n a_n q^n.$$

It raises weight by $p+1$, satisfies the product rule, and satisfies $\Theta(A) = 0$. Hence it induces

$$\Theta : Q_r \longrightarrow Q_{r+p+1}.$$

Proposition 4.5 (Theta adjointness). *If $x \in Q_r$ and $y \in Q_{P+1-p-r}$, then*

$$\langle \Theta x, y \rangle_{r+p+1} = -\langle x, \Theta y \rangle_r.$$

More generally, for $m \geq 0$ and $y \in Q_{P+2-r-m(p+1)}$,

$$\langle \Theta^m x, y \rangle_{r+m(p+1)} = (-1)^m \langle x, \Theta^m y \rangle_r.$$

Proof. The two products in the first formula have weight $P+2$, so both traces are defined. The product rule gives

$$(\Theta x)y + x(\Theta y) = \Theta(xy).$$

Notice that xy has weight

$$P+1-p = (p-1)^2.$$

Let a be the canonical section on the Igusa curve, normalized by $a^{p-1} = A$. The differential attached to $\Theta(xy)$ is $\Theta(xy)/a^P = \Theta(xy)/A^p$ under Kodaira–Spencer, and it is exact; this is the geometric description of theta [6, Proposition 5.8] (equivalently, it follows locally by differentiating the function attached to xy). The residues of this differential at the supersingular points are

exactly the local components of the Grothendieck trace in Proposition 4.4; this is the residue form of duality for the Cartier divisor cut out by A^p . An exact differential has zero residue, so $\text{Tr}_Z(\Theta(xy)) = 0$. The product rule now gives the first identity. Induction on m gives the second identity, with the weight of y decreasing by $p+1$ at each step.

For completeness, the nonreduced trace can be seen locally. At a point $s \in S$, choose a parameter t for which A is a unit times t . Then Z has local ring $k(s)[t]/(t^p)$, and the Grothendieck trace is the coefficient of t^{p-1} , followed by $\text{Tr}_{k(s)/\mathbb{F}_p}$, up to a common unit fixed by the chosen trivialization. The coefficient of t^{p-1} in the derivative of a polynomial modulo t^p is zero: it could only come from the derivative of t^p , whose coefficient is $p = 0$. This also shows that the calculation is independent of the chosen representative modulo t^p . Thus the local trace of the exact differential vanishes on the full thickening pS , not merely on its reduced support. $\square$

The weight relation relevant to the Tate complement contains a further shift of $p(p+1)$. Let

$$B := E_{p+1} \pmod{p} \in H^0(X, \omega^{p+1}).$$

Lemma 4.6 (The Eisenstein bridge is a unit). *The forms A and B have no common zero. Hence $B|_Z$ is a unit and multiplication by B^p gives an isomorphism*

$$B^p : Q_r \xrightarrow{\sim} Q_{r+p(p+1)}$$

for every r .

Proof. The Hasse invariant has a simple zero at every point of S . Let $\partial_k = \Theta - kE_2/12$ be the Serre derivative in weight k . The q -expansion of A is 1, so $\Theta(A) = 0$ in characteristic p . Also $E_2 \equiv E_{p+1} = B \pmod{p}$. Hence the q -expansion principle gives

$$12\partial_{p-1}A = B \quad \text{in characteristic } p.$$

At a zero of A , the connection term is a multiple of A and vanishes. Thus $\partial_{p-1}A$ is the ordinary derivative of a local equation for S , up to a nonzero change of trivialization. It is nonzero because the zero is simple. Therefore B does not vanish on S . Its restriction to the Artinian thickening Z is consequently invertible, and so is B^p . $\square$

Theorem 4.7 (Perfect Tate-complement pairing). *For every r the formula*

$$[x, z]_r := \langle x, B^{-p}z \rangle_r$$

defines a canonical perfect pairing

$$[\ , \]_r : Q_r \times Q_{P+2+p(p+1)-r} \longrightarrow \mathbb{F}_p.$$

Under this pairing theta is skew-adjoint. More precisely, if

$$z \in Q_{P+1-p+p(p+1)-r},$$

then

$$[\Theta x, z]_{r+p+1} = -[x, \Theta z]_r.$$

The same identity iterates with the sign $(-1)^m$ for Θ^m .

Proof. Lemma 4.6 identifies the second factor with Q_{P+2-r} , so perfectness follows from Proposition 4.4. Since theta is a derivation in characteristic p ,

$$\Theta(B^p) = pB^{p-1}\Theta(B) = 0, \quad \Theta(B^{-p}) = 0$$

on Z . Proposition 4.5 therefore gives

$$\begin{aligned} [\Theta x, z]_{r+p+1} &= \langle \Theta x, B^{-p} z \rangle_{r+p+1} \\ &= -\langle x, \Theta(B^{-p} z) \rangle_r \\ &= -\langle x, B^{-p} \Theta z \rangle_r = -[x, \Theta z]_r. \end{aligned}$$

Iteration proves the final assertion. $\square$

Lemma 4.8 (Why the undivided p -fold bridge fails). *On every Q_r one has $\Theta^p = 0$.*

Proof. Fermat's congruence and the q -expansion principle give the standard identity

$$\Theta^p = A^{p+1} \Theta$$

as operators on modular forms in characteristic p : the two sides have the same weight and the same q -expansion. In the target Hasse quotient,

$$A^{p+1} \Theta x = A^p (A \Theta x) = 0.$$

$\square$

Now let

$$w_- = k + 2i - P, \quad w_+ = k + 2i^\dagger + P, \quad i + i^\dagger = p^2 + 1 - k.$$

Then

$$w_- + w_+ = 2p^2 + 2 = P + 2 + p(p + 1),$$

and therefore

$$[\ , \]_{w_-} : Q_{w_-} \times Q_{w_+} \longrightarrow \mathbb{F}_p$$

is a perfect pairing under which theta is skew-adjoint. This is the pairing at the two candidate-weight indices. The factor B^p is exactly the weight- $p(p + 1)$ factor that occurs in the proved leading-block formula for the $D = 2$ obstruction. Serre duality alone would pair weights summing to $P + 2$; the Eisenstein bridge supplies the required extra shift. The tempting replacement by Θ^p is zero by Lemma 4.8.

There is an important indexing distinction. The obstruction to lowering to a candidate weight w naturally lies in Q_{w+P} , not Q_w ; this is proved in Proposition 4.9. Thus the pairing above is a correct structural pairing, but it is not yet the pairing of the two actual Bockstein target spaces. The latter is constructed, with the correct shift, in Theorem 4.12. This distinction prevents the constant sum of the candidate weights from being mistaken for Bockstein compatibility.

4.3 Geometric realization of the weight-lowering obstruction

Retain the notation of the Hasse-duality subsection. In particular, $P = p(p - 1)$, $S = \text{div}(A)$, $Z = pS$, and $Q_r = H^0(Z, \omega^r|_Z)$. Let $\tilde{A} = E_{p-1}$ be the normalized integral lift of the Hasse invariant. Its q -expansion is congruent to 1 modulo p , and therefore

$$\tilde{A}^p \equiv 1 \pmod{p^2} \tag{2}$$

as a q -expansion.

Proposition 4.9 (The Bockstein class is a first Hasse jet). *Let g be a divided congruence modulo p^2 . Suppose that*

1. *g has a representative $G \in M_{w+P}(\mathbb{Z}_{(p)})$ modulo p^2 ; and*

2. $g \bmod p$ has a representative $h \in M_w(\mathbb{Z}_{(p)})$.

Then

$$\Gamma_w(g) := \left(\frac{G - \tilde{A}^p h}{p} \bmod p \right) \Big|_Z \in Q_{w+P}$$

is independent of all choices. Moreover,

$$\Gamma_w(g) = 0 \iff g \text{ has a representative in } M_w(\mathbb{Z}_{(p)}) \text{ modulo } p^2.$$

Under the q -expansion identification, $\Gamma_w(g)$ is the geometric image of the divided-congruence class $\beta_w(g)$ from Lemma 4.2.

Proof. The numerator is divisible by p : both terms have weight $w + P$, and their q -expansions agree modulo p because $\tilde{A}^p \equiv 1 \pmod{p}$. Thus the displayed quotient reduces to a form of weight $w + P$ modulo p .

Changing G without changing its q -expansion modulo p^2 changes the quotient by zero modulo p , by the saturated q -expansion principle. If h' is another choice, the integral echelon argument used in Lemma 4.2 gives $h' - h = pm$ for some $m \in M_w$. The two quotients then differ modulo p by

$$-A^p m \in A^p M_w.$$

Its restriction to Z is zero. This proves independence.

The restriction sequence for the Cartier divisor Z is

$$0 \longrightarrow \omega^w \xrightarrow{A^p} \omega^{w+P} \longrightarrow \omega^{w+P}|_Z \longrightarrow 0.$$

Exactness at $H^0(X, \omega^{w+P})$ says that the kernel of restriction is exactly $A^p M_w$; no surjectivity assertion is needed here. Hence $\Gamma_w(g) = 0$ exactly when

$$\frac{G - \tilde{A}^p h}{p} \equiv A^p m \pmod{p}$$

for some $m \in M_w$. In that case

$$G \equiv \tilde{A}^p(h + pm) \pmod{p^2}.$$

By (2), the q -expansion of the right side is that of $h + pm$ modulo p^2 , so $h + pm$ is a weight- w representative of g . The converse follows by reversing the argument. Finally, before restriction to Z the displayed quotient has the same q -expansion as $(g - h)/p$ modulo p . The restriction map induces an injection

$$M_{w+P}(\mathbb{F}_p)/A^p M_w(\mathbb{F}_p) \hookrightarrow Q_{w+P},$$

and $\Gamma_w(g)$ is the image of the divided-congruence class under this map. This proves the last assertion. $\square$

The index shift in Proposition 4.9 is essential: the obstruction to lowering to weight w belongs to Q_{w+P} , not to Q_w .

Lemma 4.10 (Theta commutes with the integral Hasse inclusion). *For every integral q -series h and every $m \geq 0$,*

$$\theta^m(\tilde{A}^p h) \equiv \tilde{A}^p \theta^m(h) \pmod{p^2}.$$

Proof. Write $\tilde{A} = 1 + pa(q)$. Then $\theta \tilde{A}$ is divisible by p , and hence

$$\theta(\tilde{A}^p) = p \tilde{A}^{p-1} \theta \tilde{A} \equiv 0 \pmod{p^2}.$$

The product rule gives the assertion for $m = 1$, and iteration proves it for every m . $\square$

Lemma 4.11 (A horizontal unit of the required weight). *After pullback to a compactified modular curve with auxiliary full level 3, there is a form*

$$F \in H^0(X, \omega^{3p-1})$$

whose restriction to S is nowhere zero. Consequently

$$U := F^p|_Z \in H^0(Z, \omega^{p(3p-1)}|_Z)$$

is a unit and $\Theta(U) = 0$.

Proof. Multiplication by A gives

$$0 \longrightarrow \omega^{2p} \xrightarrow{A} \omega^{3p-1} \longrightarrow \omega^{3p-1}|_S \longrightarrow 0.$$

Serre duality and Kodaira–Spencer give

$$H^1(X, \omega^{2p})^\vee \simeq H^0(X, \omega^{2-2p}(-C)) = 0.$$

The restriction map onto S is therefore surjective. Since S is a finite reduced scheme, its coordinate ring is a product of finite fields, and every line bundle on it has a global generator. Lift such a generator to F . At every point of Z , its reduction modulo the nilpotent ideal is a generator; Nakayama’s lemma therefore shows that F generates on Z . Its p -th power is a unit of weight $p(3p-1)$, and the product rule in characteristic p gives $\Theta(F^p) = 0$. $\square$

Now put

$$w_- = k + 2i - P, \quad w_+ = k + 2i^\dagger + P,$$

and define the actual geometric Bockstein weights

$$r_- := w_- + P = k + 2i, \quad r_+ := w_+ + P = k + 2i^\dagger + 2P.$$

They satisfy

$$r_- + r_+ = P + 2 + p(3p-1). \tag{3}$$

Theorem 4.12 (Auxiliary-level natural Bockstein pairing). *Choose F as in Lemma 4.11 and put $W = p(3p-1)$. For every integer r , the formula*

$$[x, z]_{r,F}^{\text{nat}} := \langle x, U^{-1}z \rangle_r$$

defines a perfect pairing

$$Q_r \times Q_{P+2+W-r} \longrightarrow \mathbb{F}_p.$$

If $x \in Q_r$ and $z \in Q_{P+2+W-r-(p+1)}$, then the correctly typed adjointness identity is

$$[\Theta x, z]_{r+p+1,F}^{\text{nat}} = -[x, \Theta z]_{r,F}^{\text{nat}}.$$

Taking $r = r_-$ gives a perfect pairing $Q_{r_-} \times Q_{r_+} \rightarrow \mathbb{F}_p$. The construction is not canonical in F , but every choice is perfect and detects the zero class. Pullback to auxiliary level is faithful, so a level-one Bockstein vanishes if and only if its pulled-back class vanishes.

Proof. The unit U has weight W . Its inverse identifies $Q_{P+2+W-r}$ with Q_{P+2-r} . Perfectness therefore follows from Proposition 4.4. Since $\Theta(U) = \Theta(U^{-1}) = 0$, Proposition 4.5 gives

$$\begin{aligned} [\Theta x, z]_{r+p+1,F}^{\text{nat}} &= \langle \Theta x, U^{-1}z \rangle_{r+p+1} \\ &= -\langle x, \Theta(U^{-1}z) \rangle_r \\ &= -\langle x, U^{-1}\Theta z \rangle_r = -[x, \Theta z]_{r,F}^{\text{nat}}. \end{aligned}$$

Equation (3) gives $r_+ = P + 2 + W - r_-$, so the specialization at r_- has the stated factors. Faithful pullback proves the final assertion. $\square$

Remark 4.13. The family of perfect pairings in Theorem 4.12 depends on the choice of the auxiliary-level unit F . The theorem uses it only to identify dual vector spaces and detect the zero class after faithful pullback. It does not assert a canonical level-one bilinear form or descend the chosen unit through the tame deck group. By contrast, the unshifted Hasse-quotient pairing in Proposition 4.4 is canonical and does descend by invariants.

For the complementary theta iterates, Proposition 4.9 now gives the correctly typed classes

$$\Gamma_{w_-}(\theta^i f) \in Q_{r_-}, \quad \Gamma_{w_+}(\theta^{i^\dagger} f) \in Q_{r_+}.$$

The remaining cross-wedge problem is no longer the construction of their ambient pairing. It is the following concrete compatibility statement: under the isomorphism $Q_{r_+} \simeq Q_{r_-}^\vee$ induced by $[\ , \]^{\text{nat}}$, the second displayed class must equal the functional obtained from the first by the integral theta-iterate construction, up to the explicit nonzero scalar in the $D = 2$ block formula. Perfectness and theta adjointness alone do not prove that identification.

Lemma 4.10 proves the Hasse-inclusion part of the required square. Theorem 6.2 gives an explicit formula for the lower minimal-weight projection. The unresolved part is its factorial block reduction and the corresponding compatibility with the complementary class.

4.4 An abstract Bockstein-adjointness criterion

The final compatibility can be reduced to a standard algebraic check. We record the check separately so that the remaining modular input is visible.

Lemma 4.14 (Adjoint Bocksteins). *Let $R = \mathbb{Z}/p^2\mathbb{Z}$. Let $C^\bullet$ and $D^\bullet$ be bounded complexes of finite free R -modules with a perfect pairing*

$$C^a \times D^{N-a} \longrightarrow R$$

for which the differentials are graded skew-adjoint. Let

$$\begin{aligned} \partial_C : H^a(C^\bullet/p) &\longrightarrow H^{a+1}(C^\bullet/p), \\ \partial_D : H^{N-a-1}(D^\bullet/p) &\longrightarrow H^{N-a}(D^\bullet/p). \end{aligned}$$

be the connecting maps from multiplication by p . Then

$$\langle \partial_C x, y \rangle = (-1)^{a+1} \langle x, \partial_D y \rangle.$$

Proof. Choose cocycle lifts $\tilde{x}$ and $\tilde{y}$ over R . Write $d\tilde{x} = px_1$ and $d\tilde{y} = py_1$. Graded skew-adjointness gives

$$\langle d\tilde{x}, \tilde{y} \rangle = (-1)^{a+1} \langle \tilde{x}, d\tilde{y} \rangle.$$

Divide by p and reduce modulo p . The resulting classes of x_1 and y_1 are the two connecting maps, which proves the formula. $\square$

To apply Lemma 4.14 here, one must construct a self-dual two-term complex whose connecting maps are exactly the geometric classes $\Gamma_{w_-}(\theta^i f)$ and $\Gamma_{w_+}(\theta^{i^\dagger} f)$. Proposition 4.9 supplies the target spaces, and Theorem 4.12 supplies the duality. The unresolved input is commutativity of the integral theta-iterate square with the two minimal-weight projections. Lemma 4.10 proves commutativity with the E_{p-1}^p weight-raising maps. What remains is the projection term introduced by integral theta modulo p^2 , including its unit scalar. Verifying that term would turn the abstract lemma into the desired cross-wedge theorem.

5 Direct calculation in the lower subwedge

6 A direct factor-filtration route to the open theorem

The geometric pairing isolates the correct obstruction spaces, but the auxiliary horizontal unit in the natural pairing is noncanonical. For the vanishing theorem there is a more concrete route: calculate the lower Bockstein directly from the E_2 expansion, in the style of [1].

Write $i = np + i'$ and put

$$q_1(n, i') = i'^2 + (k-1)i' - (n+1)(n+2) \in \mathbb{F}_p, \quad X = E_2/12 \pmod{p}.$$

For a $D = 0$ cell let

$$w_- = k + 2i - P, \quad \mathcal{B}_i = \Gamma_{w_-}(\theta^i f) \in Q_{k+2i}.$$

On the $D = 0$ stratum used here, the integral theta expansion of [1, 2] gives a representative of $\theta^i f$ in $M_{k+2i}(\mathbb{Z}_{(p)})$ modulo p^2 . Thus Proposition 4.9 applies and the displayed class is well defined. The shifted leading-block candidate is the image in the same quotient of

$$\mathcal{L}_i = q_1(n, i') X^p \theta^{i-p} f. \quad (4)$$

Proposition 6.1 (Finite exact audit). *In the current Sturm-safe record of 692 ordinary, non-degenerate $D = 0$ - $D = 2$ complementary pairs, the following statements hold exactly.*

1. *Away from the three affine lines*

$$i' = n + 2, \quad i' + n = 2p - k - 1, \quad i' + n = 2p - k, \quad (5)$$

one has $\mathcal{B}_i = \mathcal{L}_i$ in all 458 cells.

2. *On the three lines in (5), the equality fails in all 234 cells, but $\mathcal{B}_i$ is nonzero in every cell.*
3. *The 20 vanishing classes occur off the three lines and are exactly the cells with $q_1(n, i') = 0$.*
4. *On the line $i' = n + 2$, all 78 tested residuals satisfy*

$$\mathcal{B}_i - \mathcal{L}_i = c_{p,k,n} X^{2n-2p+k+3} \theta^{p-k} f \quad \text{in } Q_{k+2i}, \quad (6)$$

for an explicitly computed scalar $c_{p,k,n} \in \mathbb{F}_p^\times$.

5. *Put $m = p - 2 - n$. On the line $i' + n = 2p - k - 1$, all 78 tested residuals lie in the explicit triangular span*

$$\left\langle X^a \theta^{2p-k-1-a} f : p-k \leq a \leq p-k+2m+1; X^p \theta^{p-k-1} f \right\rangle. \quad (7)$$

On the adjacent line $i' + n = 2p - k$, all 78 tested residuals lie in

$$\left\langle X^a \theta^{2p-k-a} f : p-k+1 \leq a \leq p-k+2m+3; X^p \theta^{p-k} f \right\rangle. \quad (8)$$

Let V_m^- and V_m^+ denote the spans in (7) and (8), respectively, after omitting the displayed X^p endpoint. In every tested cell the endpoint class is independent of the corresponding $V_m^\pm$, and

$$\mathcal{B}_i - \mathcal{L}_i \equiv -(m+1) X^p \theta^{p-k-1} f \pmod{V_m^-}, \quad i' + n = 2p - k - 1, \quad (9)$$

$$\mathcal{B}_i - \mathcal{L}_i \equiv -(m+1) X^p \theta^{p-k} f \pmod{V_m^+}, \quad i' + n = 2p - k. \quad (10)$$

Consequently every one of these 156 residuals is nonzero.

These statements are finite computations, not assertions for arbitrary p , k , and f .

Proof. The direct class is obtained by reducing $\theta^i f$ modulo p^2 by the echelon basis of M_{w_-} , dividing the canonical remainder by p , and then reducing modulo p . The candidate in (4) is reduced by the same basis modulo p . Full Sturm-safe comparison gives 458 equal and 234 unequal classes. The unequal cells are classified by (5) with confusion counts

$$\text{TP} = 234, \quad \text{FP} = 0, \quad \text{FN} = 0, \quad \text{TN} = 458.$$

The direct remainder vanishes in precisely 20 cells; each has $q_1 = 0$. The boundary residuals are nonzero. A second full-remainder comparison tests (6) in all 78 cells on $i' = n + 2$; every residual is proportional to the displayed block and every scalar is nonzero. This verifies the proposition over the stated record. Finally, exact Gaussian elimination over $\mathbb{F}_p$ on the complete remainders places every residual on the other two lines in the spans (7) and (8). The restricted echelon solution has support equal to its column rank in every case. Omitting the final X^p column lowers the rank by one in all 156 cases, while the coefficient of that final column is $-(m + 1)$ in every case. Since $1 \leq m < p - 1$, this is a nonzero scalar and proves the finite nonvanishing assertions (9) and (10). $\square$

Theorem 6.2 (Explicit periodic first-Hasse-jet formula). *Assume that $f \in M_k(\mathbb{Z}_{(p)})$ is ordinary, that $4 \leq k < p - 1$, and that (n, i') is an admissible $D = 0 - D = 2$ cross-wedge cell. Write*

$$n = p - 2 - m, \quad i' = p - k + m + 1 + d, \quad 0 \leq d \leq k - 2m - 1.$$

Put

$$r = p - k + d, \quad L = n + 1, \quad \ell_0 = p - k + 1.$$

Then $i = np + i' = r + L(p - 1)$.

Let $A = E_{p-1}$ and $B = E_{p+1}$ be their standard p -integral Eisenstein lifts, and, for $g \in M_v(\mathbb{Z}_{(p)})$, define

$$\mathcal{T}_v(g) = A\theta g + \frac{v}{12}(B - AE_2)g \in M_{v+p+1}(\mathbb{Z}_{(p)}). \quad (11)$$

Let $F_j(f)$ be the modified Serre derivatives, defined by

$$F_0(f) = f, \quad F_1(f) = \partial_k f, \quad F_{j+1}(f) = \partial_{k+2j} F_j(f) - \frac{j(j+k-1)}{144} E_4 F_{j-1}(f),$$

where $\partial_v g = \theta g - (v/12)E_2 g$. At the first ordinary low point, the Chida-Kaneko congruence [3] gives

$$F_{\ell_0}(f) \in M_{k+2\ell_0}(\mathbb{Z}_{(p)}), \quad F_{\ell_0}(f) \equiv \theta^{\ell_0} f \pmod{p}.$$

Define a modular lift G_r of $\theta^r f$ as follows. If $r < \ell_0$, start with $G_0 = f$ and apply (11) r times. If $r \geq \ell_0$, start with $G_{\ell_0} = F_{\ell_0}(f)$ and apply (11) $r - \ell_0$ times. Its weight is

$$v_r = \begin{cases} k + r(p + 1), & r < \ell_0, \\ k + r(p + 1) - \ell_0(p - 1), & r \geq \ell_0. \end{cases}$$

If

$$a = \frac{w_- - v_r}{p - 1} = \begin{cases} k - 2m - 2, & d = 0, \\ p - 2m - d - 1, & d \geq 1, \end{cases}$$

then $a \geq 0$ and

$$H_i = A^a G_r \in M_{w_-}(\mathbb{Z}_{(p)}), \quad H_i \equiv \theta^i f \pmod{p}. \quad (12)$$

In $\mathbb{F}_p[[q]]$, put

$$\begin{aligned}\alpha &= \frac{A-1}{p} \pmod{p}, \\ \Phi_r(f) &= \frac{\theta^{r+p-1}f - \theta^r f}{p} \pmod{p}, \\ \eta_r(f) &= \frac{G_r - \theta^r f}{p} \pmod{p}.\end{aligned}$$

These quotients are integral. The lower Bockstein is the class

$$\boxed{\mathcal{B}_i = [L\Phi_r(f) - \eta_r(f) - a\alpha\theta^r f]_{Q_{k+2i}}.} \quad (13)$$

Thus the minimal-weight projection correction is an explicit first-Hasse-jet operator; it is not an unspecified choice of characteristic-zero lift.

Proof. The Serre derivative identity rewrites (11) as

$$\mathcal{T}_v(g) = A\left(\theta g - \frac{v}{12}E_2 g\right) + \frac{v}{12}Bg,$$

so it is modular of weight $v + p + 1$. The congruences $A \equiv 1$ and $B \equiv E_2 \pmod{p}$ give $\mathcal{T}_v(g) \equiv \theta g \pmod{p}$. Together with the Chida–Kaneko low-point congruence, induction gives $G_r \equiv \theta^r f \pmod{p}$ and the displayed formula for v_r . Substitution of $r = p - k + d$ and the two possible filtration branches gives the stated value of a . The cross-wedge inequalities make both formulas for a nonnegative. Hence (12) follows.

For a Fourier index j prime to p , write $j^{p-1} = 1 + pq_p(j) \pmod{p^2}$. Since $i = r + L(p-1)$,

$$j^i = j^r(1 + pLq_p(j)) \pmod{p^2}.$$

If $p \mid j$, then $r \geq p - k \geq 2$, and both sides of the corresponding divided congruence vanish modulo p . Coefficientwise, therefore,

$$\frac{\theta^i f - \theta^r f}{p} \equiv L\Phi_r(f) \pmod{p}. \quad (14)$$

Also $A = 1 + p\alpha \pmod{p^2}$ and $G_r = \theta^r f + p\eta_r(f) \pmod{p^2}$, whence

$$H_i = A^a G_r \equiv \theta^r f + p(\eta_r(f) + a\alpha\theta^r f) \pmod{p^2}.$$

Subtracting this identity from (14), dividing by p , and passing to the lower Hasse quotient proves (13). $\square$

The first Hasse jet in (13) is itself recursive. Put

$$C = \frac{B - AE_2}{p} \pmod{p}.$$

Whenever $G_{t+1} = \mathcal{T}_{v_t}(G_t)$, direct expansion modulo p^2 gives

$$\eta_{t+1} = \theta\eta_t + \alpha\theta^{t+1}f + \frac{v_t}{12}C\theta^t f. \quad (15)$$

The initial jet is $\eta_0 = 0$ on the first branch and $(F_{\ell_0}(f) - \theta^{\ell_0}f)/p$ on the second. Equations (13) and (15) therefore reduce the remaining block identities to a finite symbolic calculation in three explicit divided q -series.

Lemma 6.3 (Unit-block endpoint coefficient). *Suppose that $i = np + i'$ lies on either of the two triangular lines in (5), and put $m = p - 2 - n$. Define*

$$\kappa_A = \frac{1}{p} \binom{i}{i-p} \prod_{r=i-p+k}^{i+k-1} r \pmod{p}.$$

Then

$$\kappa_A = -n(n+1) = -(m+1)(m+2) \pmod{p},$$

and the coefficient of the $X^p \theta^{i-p} f$ unit block after subtracting the shifted coefficient $q_1(n, i')$ is

$$\kappa_A + i(i+k-1) - q_1(n, i') = -2(m+1) \pmod{p}. \quad (16)$$

Proof. The quotient defining κ_A is an integer because the product contains a factor divisible by p . Lucas's theorem gives $\binom{i}{i-p} = \binom{i}{p} \equiv n \pmod{p}$. On the first triangular line one has $i' = p - k + m + 1$, and on the second one has $i' = p - k + m + 2$. In either case the product defining κ_A consists of p consecutive integers and its unique multiple of p is $(n+1)p$. After division by p , the remaining factors run through $\mathbb{F}_p^\times$. Wilson's theorem therefore gives

$$\frac{1}{p} \prod_{r=i-p+k}^{i+k-1} r \equiv -(n+1) \pmod{p},$$

which proves the formula for κ_A . Since $i(i+k-1) \equiv i'(i'+k-1)$ and

$$q_1(n, i') = i'(i'+k-1) - (n+1)(n+2),$$

their difference is

$$-n(n+1) + (n+1)(n+2) = 2(n+1) = -2(m+1) \pmod{p}.$$

□

Proposition 6.4 (Uniform separation of the triangular endpoint). *Assume that f is ordinary at p , that $4 \leq k < p-1$, and that a $D = 0-D = 2$ cross-wedge cell lies on one of the two triangular lines in (5). Put $m = p - 2 - n$. Then*

$$1 \leq m \leq \frac{k-4}{2}.$$

On the line $i' + n = 2p - k - 1$, the class

$$X^p \theta^{p-k-1} f$$

is not in $V_m^- + M_{w-}(\mathbb{F}_p)$. On the line $i' + n = 2p - k$, the class

$$X^p \theta^{p-k} f$$

is not in $V_m^+ + M_{w-}(\mathbb{F}_p)$. In particular, the endpoint classes in (9) and (10) are independent of the preceding triangular spans for every admissible p, k, m , not only in the finite audit.

Proof. Let $\ell_0 = p - k + 1$ be the first low point of the ordinary mod- p theta cycle. For $0 \leq t < p$, its filtration is

$$\omega_p(\theta^t f) = \begin{cases} k + t(p+1), & t < \ell_0, \\ k + t(p+1) - (p-k+1)(p-1), & t \geq \ell_0. \end{cases}$$

By Lemma 4.6, the reduction of E_{p+1} is a unit along the Hasse divisor. Since $X = E_2/12 \equiv E_{p+1}/12 \pmod{p}$ and 12 is a unit, multiplication by X does not change the order of Hasse divisibility. Therefore

$$\omega_p(X^a \theta^t f) = \omega_p(\theta^t f) + a(p+1).$$

On the first triangular line, every generator of V_m^- has $a+t = 2p-k-1$ and

$$p-2m-2 \leq t \leq p-1.$$

The cross-wedge inequalities give $2m+2 \leq k-2$, hence $t \geq p-k+2 > \ell_0$. All these generators therefore have exact filtration

$$F_-^{\text{old}} = k + (2p-k-1)(p+1) - (p-k+1)(p-1) = p^2 + p - k.$$

The endpoint has $a = p$ and $t = p-k-1 < \ell_0$, so its exact filtration is

$$F_-^{\text{end}} = k + (2p-k-1)(p+1) = p(2p+1-k) - 1.$$

Thus

$$F_-^{\text{end}} - F_-^{\text{old}} = (p-k+1)(p-1) > 0.$$

Moreover a direct substitution in $w_- = k + 2i - p(p-1)$ gives

$$F_-^{\text{old}} - w_- = 2(m+1)(p-1) > 0.$$

On the second triangular line, every generator of V_m^+ has $a+t = 2p-k$ and

$$p-2m-3 \leq t \leq p-1.$$

Here $2m+3 \leq k-1$, so again $t \geq \ell_0$. Their common exact filtration is

$$F_+^{\text{old}} = k + (2p-k)(p+1) - (p-k+1)(p-1) = p^2 + 2p - k + 1.$$

The endpoint has $a = p$ and $t = p-k < \ell_0$, and hence

$$F_+^{\text{end}} = k + (2p-k)(p+1) = p(2p+2-k).$$

Again

$$F_+^{\text{end}} - F_+^{\text{old}} = (p-k+1)(p-1) > 0, \quad F_+^{\text{old}} - w_- = (2m+3)(p-1) > 0.$$

If an endpoint belonged to the sum of the preceding span and M_{w_-} , its filtration would be at most the larger of w_- and the filtrations of the preceding generators. The strict inequalities above contradict the exact endpoint filtration. This proves both assertions. $\square$

The exact audit further separates the endpoint in (9)–(10). The triangular B -block sum has zero distinguished-endpoint coefficient, while the explicit first-Hasse-jet correction in (13) has coefficient $m+1$ in all 156 tested cells. Together with (16), this gives

$$-2(m+1) + 0 + (m+1) = -(m+1).$$

Thus the open uniform statement on these two lines is now the associated-graded simplification

$$\Pi_i \equiv (m+1)X^p \theta^{p-k-\varepsilon} f \pmod{V_m^\varepsilon}, \quad \varepsilon = \begin{cases} 1, & i' + n = 2p-k-1, \\ 0, & i' + n = 2p-k, \end{cases} \quad (17)$$

where $V_m^1 = V_m^-$ and $V_m^0 = V_m^+$. The independence of the displayed endpoint is Proposition 6.4; the first-Hasse-jet formula itself is Theorem 6.2. Only its simplification to the displayed coefficient remains open.

There is a parallel uniform statement on the third line. It does not determine the scalar in (6), but it proves that the displayed block is a genuine, nonzero quotient direction.

Proposition 6.5 (The third-line block is nonzero). *Assume the hypotheses of Proposition 6.4 and suppose that $i' = n + 2$. Put $m = p - 2 - n$. Then $i' = p - m$, $1 \leq m \leq (k - 4)/2$, and*

$$Y_{p,k,m} := X^{k-2m-1} \theta^{p-k} f$$

has exact filtration

$$\omega_p(Y_{p,k,m}) = w_- + (p - 1).$$

In particular, its image in the lower Hasse quotient is nonzero. Also

$$q_1(n, i') = -km \not\equiv 0 \pmod{p}.$$

Proof. The equality $i' = n + 2$ and the definition $m = p - 2 - n$ give $i' = p - m$. The cross-wedge inequalities give $1 \leq m \leq (k - 4)/2$. Since $p - k$ lies strictly before the first low point $p - k + 1$ of the ordinary theta cycle,

$$\omega_p(\theta^{p-k} f) = k + (p - k)(p + 1).$$

Multiplication by X^{k-2m-1} raises exact filtration by $(k - 2m - 1)(p + 1)$. Hence

$$\omega_p(Y_{p,k,m}) = k + (p - 2m - 1)(p + 1).$$

On this line

$$i = (p - 2 - m)p + (p - m) = p^2 - p - m(p + 1),$$

so

$$w_- = k + 2i - p(p - 1) = k + p^2 - p - 2m(p + 1).$$

Subtracting gives $\omega_p(Y_{p,k,m}) - w_- = p - 1$. Thus $Y_{p,k,m}$ cannot lie in $M_{w_-}(\mathbb{F}_p)$ and its quotient class is nonzero. Finally, direct substitution of $n = p - 2 - m$ and $i' = p - m$ in q_1 gives $q_1 = -km$. Both k and m lie strictly between 0 and p , so this scalar is nonzero. $\square$

If the residual belongs to the line spanned by $Y_{p,k,m}$, its scalar in (6) is uniquely determined. Under this one-line hypothesis it can be read in canonical quotient coordinates as

$$c_{p,k,m}(f) = \frac{\lambda(\mathcal{B}_i - \mathcal{L}_i)}{\lambda(Y_{p,k,m})}, \tag{18}$$

where λ is any linear functional nonzero on the one-dimensional line spanned by $Y_{p,k,m}$. Because the residual is then a scalar multiple of $Y_{p,k,m}$, the ratio is independent of λ . Exact reconstruction of the computed values identifies the following closed candidate:

$$c_{p,k,m}^* = -m(k - 2m - 2)! \pmod{p}. \tag{19}$$

Proposition 6.6 (The closed third-line scalar is a unit). *For every admissible p, k, m , the value in (19) belongs to $\mathbb{F}_p^\times$.*

Proof. The cross-wedge inequalities give

$$1 \leq m \leq \frac{k - 4}{2}, \quad 2 \leq k - 2m - 2 \leq k - 4 < p.$$

Thus neither m nor any factor in $(k - 2m - 2)!$ is divisible by p . Their product is nonzero modulo p . $\square$

The finite evidence for this stronger statement is broader than the discovery table. A blind extension through $p = 61$ checked 464 third-line cells and found no zero scalar, no failure of the one-block shape. A separate weight-24 test, where the cusp-form space is two-dimensional, checked 1,860 cells from 186 ordinary forms and linear combinations through $p = 43$. In both audits every computed scalar was exactly (19); there were no mismatches. The scalar was also independent of the form for fixed p, k, m . These exact computations identify the formula and strongly test it; they do not replace the symbolic block reduction below.

The generic identity and the third-line formula now have one common target. For the parameters of Theorem 6.2, put

$$Z_d = X^d \theta^{p-k} f.$$

A direct filtration calculation gives

$$\omega_p(Z_d) - w_- = (d - k + 2m + 2)(p - 1). \quad (20)$$

Consequently $Z_d \in M_{w_-}(\mathbb{F}_p)$ for $d \leq k - 2m - 2$, whereas $Z_{k-2m-1} = Y_{p,k,m}$ is the nonzero direction of Proposition 6.5. It is therefore enough to prove the following single symbolic reduction.

Conjecture 6.7 (Factorial block reduction). *For $2 \leq d \leq k - 2m - 1$, the explicit first-Hasse-jet class satisfies*

$$\mathcal{B}_i - \mathcal{L}_i = -m(d - 1)! [Z_d] \quad \text{in } Q_{k+2i}. \quad (21)$$

For $2 \leq d \leq k - 2m - 2$, equation (20) makes the right side of (21) zero, proving the generic shifted identity. At $d = k - 2m - 1$, the same equation becomes (6) with the explicit scalar (19), which is a unit by Proposition 6.6. Thus one factorial reduction proves both the generic identity and the complete third-line assertion.

Theorem 6.2 replaces the formerly unspecified minimal-weight projection compatibility square by an exact operator formula. What remains is the formal reduction of (13) to (21), together with the two triangular endpoint coefficients. The audit identifies those associated-graded coefficients as the unit $-(m + 1)$. Proposition 6.4 already supplies endpoint independence; it must not be counted among the remaining conjectural inputs.

7 Interpretation of the experiments

The main experiment finds a clear structure. One quadratic controls the interior of the cross-wedge. Three affine boundary lines contain extra quotient directions. The same failures remain when the Fourier cutoff is increased. They are therefore separate filtration cases, not truncation errors.

The proofs explain several parts of this pattern for all admissible parameters. The first-Hasse-jet formula explains why the $D = 2$ block fails in the new wedge. Exact filtration shows that the three boundary directions survive in the quotient. The perfect Hasse-quotient pairing and theta skew-adjointness connect the two obstructions. These theorems do not give every boundary coefficient. The missing steps are the factorial block reduction in Conjecture 6.7 and two endpoint identities.

The evidence is strongest on the third boundary line. The one-block shape was first observed in 78 cells, then tested blindly in 464 further cells, and finally tested on 186 ordinary forms in a two-dimensional cusp-form space. All 2,324 tests returned the same form-independent factorial scalar. The formula is a precise and well-tested conjecture. The finite tests do not replace the missing symbolic proof.

There are three limits. First, the main cross-wedge table ends at $p = 43$, although the separate third-line test reaches $p = 61$ and the symbolic index audit reaches $p = 101$. Second, the broad record uses level-one ordinary forms; supersingular and higher-level behavior is not tested. Third, the two triangular coefficient identities have exact finite support and proved

endpoint independence, but their uniform block-basis simplification is still open. These limits give clear next tests and proof problems.

8 Tate-dual obstruction classes

Put $P = p(p-1)$ and let $i = np + i'$ be in the nondegenerate band. Define

$$i^\dagger = p^2 + 1 - k - i = (p-2-n)p + (2p+1-k-i').$$

On every admissible cell considered here, the mod- p^2 filtration has the same weight class as $k+2i$. We therefore write

$$\omega_{p^2}(\theta^i f) = k + 2i + \delta(\theta^i f)P.$$

This equation defines the integer $\delta(\theta^i f)$. Thus $\delta = -1$ means that one additional full Hasse drop occurs below $k+2i$, while $\delta = 0$ means that it does not. Then

$$i + i^\dagger = p^2 + 1 - k, \quad (k+2i) + (k+2i^\dagger) = 2p^2 + 2.$$

Moreover, if $c(x) = x^2 + (k-1)x$, then $c(2p+1-k-i') \equiv c(i') \pmod{p}$, while

$$(p-2-n)(p-1-n) \equiv (n+1)(n+2) \pmod{p}.$$

The lower Bockstein class $\beta_w(g)$ and its lift-independence are given by Lemma 4.2. In particular,

$$\beta_w(g) = 0 \iff \omega_{p^2}(g) \leq w$$

for an admissible candidate weight w .

The perfect Hasse-quotient pairing and theta adjointness are proved in Propositions 4.4 and 4.5. Put

$$w_- = k + 2i - P, \quad w_+ = k + 2i^\dagger + P.$$

Proposition 4.9 places the two actual weight-lowering obstructions in

$$Q_{r_-} \quad \text{and} \quad Q_{r_+}, \quad r_- = w_- + P = k + 2i, \quad r_+ = w_+ + P = k + 2i^\dagger + 2P.$$

Their weights satisfy

$$r_- + r_+ = P + 2 + p(3p-1).$$

Theorem 4.12 uses a horizontal unit of weight $p(3p-1)$ to give a perfect pairing

$$[\ , \]^{\text{nat}} : Q_{r_-} \times Q_{r_+} \longrightarrow \mathbb{F}_p$$

as part of a correctly typed family under which theta is skew-adjoint. The pairing shows that the two obstruction spaces are dual. It does not by itself identify the two particular obstruction classes.

Conjecture 8.1 (Cross-wedge Bockstein compatibility). *Assume that f is ordinary at p , $4 \leq k < p-1$, and that both i and $i^\dagger$ lie in the nondegenerate band. If*

$$D(i) = 0, \quad D(i^\dagger) = 2,$$

then the two geometric Bockstein classes

$$\Gamma_{w_-}(\theta^i f) \in Q_{r_-}, \quad \Gamma_{w_+}(\theta^{i^\dagger} f) \in Q_{r_+}$$

vanish together:

$$\Gamma_{w_-}(\theta^i f) = 0 \iff \Gamma_{w_+}(\theta^{i^\dagger} f) = 0.$$

Equivalently,

$$\beta_{k+2i-P}(\theta^i f) = 0 \iff \beta_{k+2i^\dagger+P}(\theta^{i^\dagger} f) = 0.$$

The conjecture states only the intrinsic vanishing equivalence needed below. A stronger equality through a chosen auxiliary-level pairing would require an explicit comparison map and a normalization of the horizontal unit. Those data have not yet been constructed, so no such equality is claimed here.

Conjecture 8.2 (Exact $D = 0$ shifted-conic law). *Under the hypotheses of the preceding conjecture,*

$$\delta(\theta^i f) = -1 \iff i'^2 + (k-1)i' - (n+1)(n+2) \equiv 0 \pmod{p}.$$

Otherwise $\delta(\theta^i f) = 0$.

Proposition 8.3 (Conditional shifted-conic deduction). *Conjecture 8.1 implies Conjecture 8.2.*

Proof. For the dual index, the proved $D = 2$ polynomial is

$$(i'^\dagger)^2 + (k-1)i'^\dagger - n^\dagger(n^\dagger + 1).$$

The two elementary congruences above transform this into

$$i'^2 + (k-1)i' - (n+1)(n+2).$$

The $D = 2$ theorem of Ahlgren–Raum–Richter [1] says that the complementary obstruction vanishes exactly on this conic. The assumed cross-wedge compatibility transfers this vanishing to i . This gives the claimed equivalence and proves the implication. $\square$

Conjecture 8.4 (Corrected global duality). *Let $e(i) = D(i) - \delta(\theta^i f)$. Away from the central $D = 1$ plateau,*

$$e(i) = e(i^\dagger).$$

On that plateau the difference is a rank-one boundary functional supported on

$$i'^2 + (k-1)i' - (n+1)^2 \equiv 0 \pmod{p}.$$

9 Remaining symbolic identities

Theorem 6.2 constructs the minimal mod- p representative in characteristic zero and computes its normal coefficient along the Hasse divisor. A complete cross-wedge theorem now requires the factorial block reduction in Conjecture 6.7 and the two triangular associated-graded endpoint coefficients. The residual directions and their independence are already proved. The open task is to simplify the explicit operator formula in the direct block basis.

On the third line, that simplification predicts

$$\mathcal{B}_i - \mathcal{L}_i = -m(k-2m-2)!X^{k-2m-1}\theta^{p-k}f.$$

The displayed block is nonzero by Proposition 6.5, and the scalar is a unit by Proposition 6.6. Thus the unresolved point is not nonvanishing, but identification of the computed scalar with the formal block reduction for every admissible parameter set.

10 Conclusion

The exact experiments find a shifted quadratic in the interior of the next theta-cycle wedge. They also find three boundary residuals. The geometry and filtration results place these residuals in the predicted quotient directions. The first-Hasse-jet formula reduces the remaining problem to three coefficient identities. The factorial block reduction and the two triangular endpoint identities remain open. The table below separates the uniform theorems, finite results, conditional claims, and conjectures.

11 Theorem status

The table gives the status of every claim used in the proposed shifted-conic application.

Statement	Status	Scope or limitation
Bockstein realization as a first Hasse jet; perfect Hasse-quotient pairing; theta skew-adjointness	Proved uniformly	Geometric results on the admissible level cover, with descent and indexing made explicit.
Integral Hasse inclusion and explicit periodic first-Hasse-jet formula	Proved uniformly	Ordinary level-one forms with $4 \leq k < p - 1$ in the admissible cross-wedge.
Boundary-direction separation and nonvanishing of the factorial scalar candidate	Proved uniformly	Establishes the quotient directions and that $-m(k - 2m - 2)!$ is a unit; it does not identify the actual residual scalar.
Direct obstruction pattern and exact scalar agreement	Verified exactly in finite range	692 cross-wedge cells and 2,324 third-line tests, all Sturm-safe and exactly reproducible.
Factorial block reduction and two triangular endpoint coefficients	Conjectural	The remaining direct block-basis calculation.
Cross-wedge Bockstein compatibility, shifted-conic law, and corrected global duality	Conditional	Follow from the stated block and compatibility conjectures; they are not used to prove the unconditional theorems.

A Computational reproducibility

A.1 Scope and arithmetic

The code has three jobs. It rejects false formulas, finds exact boundary patterns, and checks that the explicit constructions give the canonical remainders. A finite check is never used as a proof for all primes. Every modular reduction and linear system is evaluated over $\mathbb{Z}/p^2\mathbb{Z}$ or $\mathbb{F}_p$ with integer arithmetic. The reported audits use no floating-point tolerance, random seed, or random search.

The reference environment is Python 3.13.7 with NumPy 2.4.6 and SymPy 1.14.0. From the project directory it can be installed with

```
python -m pip install -r requirements.txt
```

The deposited comma-separated files in `data/` contain the cell parameters and canonical obstruction coordinates. The implementation in `code/branchfree.py` constructs level-one echelon bases and keeps enough Fourier coefficients to pass the Sturm bound at every tested weight.

A.2 Primary checks

All four main audits can be run with

```
python run_core_audit.py
```

The script stops with a nonzero exit code if a command fails or if an expected exact summary is missing. The individual commands are given below.

The explicit lower-weight lift and periodic first-Hasse-jet formula are checked by

```
python verify_periodic_first_jet.py data/d0_obstruction_coordinates.csv
```

The expected terminal summary is

```
rows=692 checked=692 bad_weight=0 bad_mod_p=0 bad_beta=0
```

Here `bad_weight` tests the exact weight of the constructed lift, `bad_mod_p` tests its congruence with $\theta^i f$, and `bad_beta` compares the resulting divided series with the canonical echelon Bockstein.

The blind third-line extension is

```
python stress_test_l3_uniformity.py --max-p 61
```

and ends with

```
checked=464 zero=0 failed_shape=0 wrong_scalar=0
```

The independent weight-24 test is

```
python stress_test_l3_weight24.py --max-p 43
```

and ends with

```
ordinary_forms=186 checked=1860 zero=0 failed_shape=0 wrong_scalar=0
```

In both commands, `wrong_scalar` compares the computed value with $-m(k - 2m - 2)!$.

Finally,

```
python audit_index_formulas.py
```

checks the exponent decompositions, lower-weight formulas, Hasse exponents, pairing sums, and filtration gaps over all admissible symbolic parameters for primes through 101. This audit is deliberately separate from the Fourier-coefficient calculation.

A.3 Interpretation

The checks show that the programs implement the stated formulas on the deposited range. They also show that the first-Hasse-jet lift agrees with an independent canonical reduction. They do not prove Conjecture 6.7 or the two triangular endpoint coefficients for arbitrary p . Those statements remain clearly separated from the unconditional Hasse-jet and boundary-separation theorems.

Data and code availability

The public release contains the cell tables, obstruction coordinates, source code, and commands used for every computational count in this paper. It contains only research files. The permanent repository links are given in the submission record.

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